

Then, we extend Lemma C.3 to d -dimensional case.

Lemma C.6. *Let*

$$f_1(x) := \exp(\langle u, x \rangle^3 + (-2 - 3C)\langle u, x \rangle^2 + (3C^2 + 4C)\langle u, x \rangle + \ln 2) - 1$$

where $C \approx -2.15$, $u := [1, \frac{1}{d-1}, \dots, \frac{1}{d-1}, \frac{1}{d-1} + C - 1]^\top \in \mathbb{R}^d$. It holds that

$$f_1([x_1, \mathbf{1}_{d-1}^\top]^\top) = \exp(x_1^3 - 2x_1^2) - 1$$

Proof. Note that $\langle u, [x_1, \mathbf{1}_{d-1}^\top]^\top \rangle = x_1 + C$. Hence

$$\begin{aligned} f_1([x_1, \mathbf{1}_{d-1}^\top]^\top) &= \exp((x_1 + C)^3 + (-2 - 3C)(x_1 + C)^2 + (3C^2 + 4C)(x_1 + C) + \ln 2) - 1 \\ &= \exp(x_1^3 - 2x_1^2) - 1 \end{aligned}$$

where the first step follows from $\langle u, [x_1, \mathbf{1}_{d-1}^\top]^\top \rangle = x_1 + C$, and the second step follows from expanding the higher-order terms and simplifying it. $\square$

Then, we extend Lemma C.4 to d -dimensional case.

Lemma C.7. *Let $W := [w_1, w_2, \dots, w_d] \in \mathbb{R}^{d \times d}$ be a weight matrix. Let $C := -2.15$ be a constant. Let a exponential activation function $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as*

$$g(z) := \exp(z^3 + (-2 - 3C)z^2 + (3C^2 + 4C)z + \ln 2) - 1.$$

We define one layer of a neural network, denoted as f , as follows:

$$f(x; W) := [g(\langle w_1, x \rangle), g(\langle w_2, x \rangle), \dots, g(\langle w_d, x \rangle)]^\top.$$

Then there exists a weight matrix W following statements hold:

- *The function $f(x; W)$ has at least two fixed points which can be found by the fixed-point iteration, and we denoted them as p_1, p_2 .*
- *For $i \in \{1, 2\}$, there exists a constant $\epsilon_i > 0$, for any initial point $x^{(0)} \in [p_{i,1} - \epsilon_i, p_{i,1} + \epsilon_i] \times \{1\} \times \{1\}$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)}; W)$ converges to the fixed point p_i .*
- *For $i \in \{1, 2\}$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 1)$ such that for any $t \geq 2$,*

$$\|x^{(t)} - p_i\|_\infty \leq K_i^t \cdot c_i \epsilon_i.$$

Proof. The lemma holds by combining Lemma C.1, Lemma C.5, and Lemma C.6. $\square$